

WASP-2b

R-0142

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_190614 · created 2026-07-16T07:47:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458648.9215 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1604 +/- 0.0071
Transit depth (Rp/Rs)^2	2.57 +/- 0.23 [%]
Orbital Inclination (inc)	85.53 +/- 0.55
Airmass coefficient 1 (a1)	1.132 +/- 0.015
Airmass coefficient 2 (a2)	-0.103 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#3 - [517, 42]
Optimal Aperture	5.11
Optimal Annulus	9.72
Transit Duration (day)	0.0834 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1604 ± 0.0071 vs 0.1326 (deviation 3.25σ)
Duration fit vs published	0.0834 d vs 0.0737 d (deviation 1.78σ)
Mid-time O–C	-3.2 min
Depth SNR	18.7
Residual scatter	1.29 %

Light curve

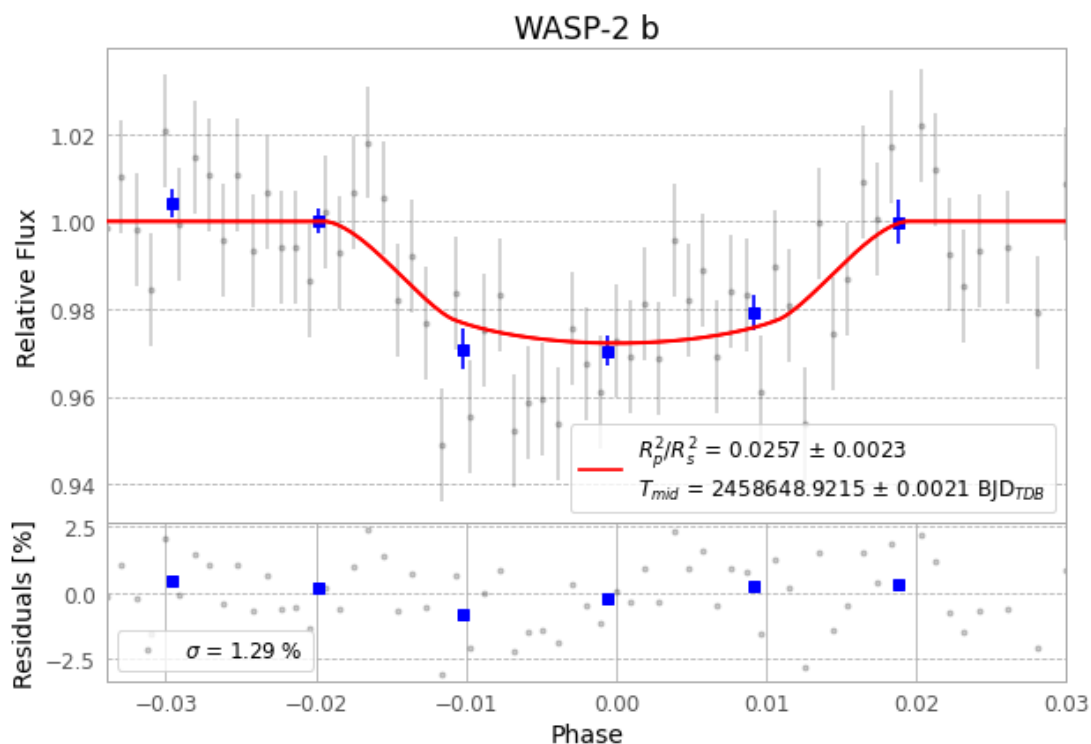


Figure 1 — FinalLightCurve_WASP-2 b_2019-06-14.png (SHA-256 2fe492bee67d89a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 104], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a9f74b8ac3223e7e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

64 FITS frames (42 MB), WASP-2190614081512.FITS → WASP-2190614113320.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-190614.log (d5c6c4ec5fa1...), FinalLightCurve_WASP-2 b_2019-06-14.png (2fe492bee67d...), FinalLightCurve_WASP-2 b_2019-06-14.csv (7815daa6b086...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 07:47Z.